Reduced Heart Rate Variability Links to Subclinical Atherosclerosis in Systemic Sclerosis and Idiopathic Inflammatory Myopathies

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Introduction

- Patients with systemic sclerosis (SSc) and idiopathic inflammatory myopathies (IIM) have increased cardiovascular (CV) risk, underestimated by conventional risk scores.
- Reduced heart rate variability (HRV), a marker of impaired vagal activity, is linked to CV morbidity and mortality.
- **Aim:** to investigate the relationship between HRV and CV risk in SSc and IIM.

Methods

- Retrospective study, patients without CV disease (**Table 1**):
  - 38 SSc (2013 ACR/EULAR criteria): 26 lcSSc, 12 dcSSc
  - 30 IIM (2017 ACR/EULAR criteria): 11 ASyS, 9 DM, 7 IMNM, 3 PM
- CV risk assessment with a resting 10-s ECG:
  - risk scores: SCORE, SCORE2, mSCORE (= SCORE × 1.5)
  - pulse wave velocity (PWV), ankle-brachial index (ABI)
  - carotid ultrasound (US): carotid intima-media thickness (CIMT), plaques, CV risk category
- HRV quantified as Root Mean Square of Successive Differences (RMSSD; marker of vagal activity) and compared with age- and sex-matched published reference values of healthy individuals[1]
- Linear regression of CV risk measures and disease activity (MYOACT) on ln(RMSSD), adjusted for age and sex

Table 1. Baseline characteristics

Mean ± SD, median (IQR) or n (%) of patients with available data

Characteristic	SSc (n=38)	IIM (n=30)
Age [years]	52.8 ± 13.2	55.2 ± 15.3
Female	35 (92%)	24 (80%)
Disease duration [years]	4.5 (1.3-12.0)	1.0 (0.7-4.0)
Interstitial lung disease	19 (50%)	15 (50%)
Glucocorticoids	10 (26%)	29 (97%)
BMI [kg/m²]	24.5 (21.7-29.6)	27.6 (23.5-30.3)
Hypertension	10 (26%)	19 (63%)
Diabetes mellitus	1 (3%)	6 (20%)
Dyslipidemia	22 (58%)	26 (87%)
Current smoking	5 (13%)	1 (3%)
SCORE [%]	1.0 (0.0-5.0)	3.5 (1.0-7.0)
SCORE2 [%]	2.9 (1.3-8.7)	5.8 (2.5-10.7)
CIMT [mm]	0.78 (0.68-0.90)	0.70 (0.60-0.80)
Carotid plaques	14 (41%)	13 (43%)
High / very high US CV risk	15 (44%)	13 (43%)
RMSSD [ms]	18.1 (11.1-24.9)	19.7 (14.2-25.5)

Figure 1. Vagal activity in patients vs. age- and sex-matched healthy controls

HC = age- and sex-matched median RMSSD from a published reference cohort of 10-s ECGs[1]. Boxplots show median and IQR.

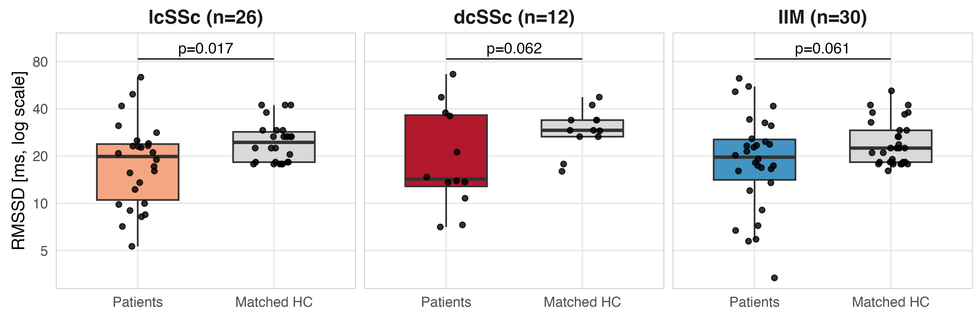
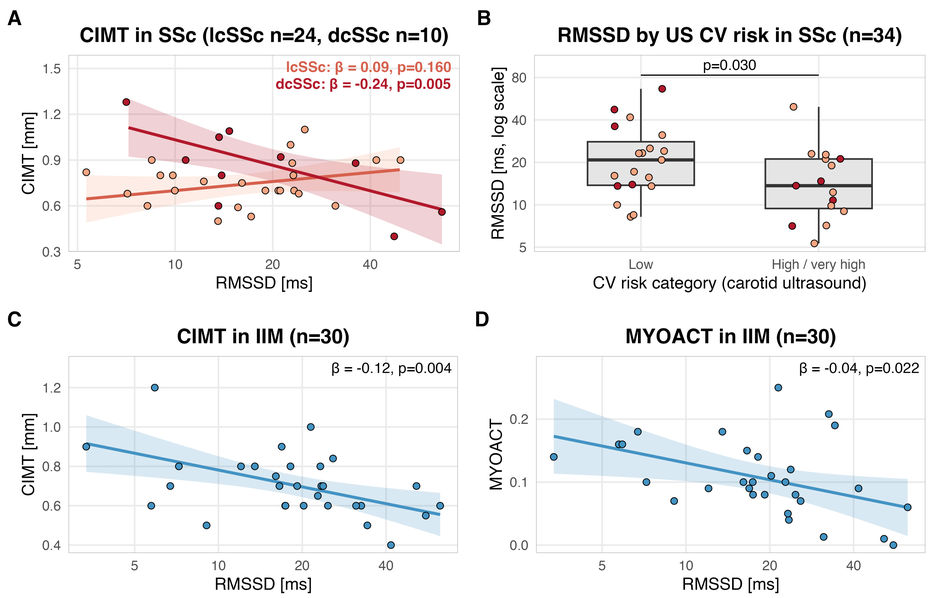


Figure 2. Lower vagal activity (RMSSD) is associated with carotid atherosclerosis and disease activity

A, C, D: linear regression adjusted for age and sex (shaded 95% CI), β per 1-unit increase in ln(RMSSD); SSc slopes differ by subtype (interaction p=0.002) orange = lcSSc, red = dcSSc. B: vagal activity stratified by CV risk category assessed by carotid ultrasound

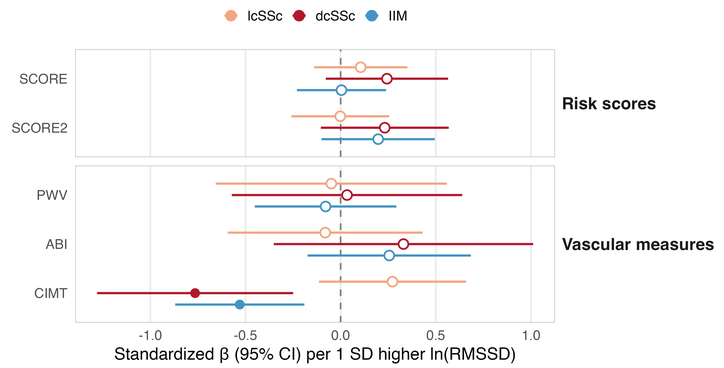


Results

- **Reduced vagal activity (Fig. 1):** RMSSD was lower in SSc than in matched HC (median 18.1 vs. 26.6 ms; p=0.001), in lcSSc (p=0.017) and numerically in dcSSc (p=0.062). In IIM, RMSSD was numerically reduced (19.7 vs. 22.5 ms; p=0.061).
- **Subclinical atherosclerosis (Fig. 2A-C):** lower ln(RMSSD) was associated with higher CIMT in dcSSc (n=10; β=−0.24, p=0.005) and IIM (β=−0.12, p=0.004), not in lcSSc (β=0.09, p=0.160). SSc patients at high ultrasound CV risk had lower RMSSD (13.7 vs. 20.8 ms; p=0.030).
- **lcSSc vs. dcSSc:** lcSSc patients were older (56 vs. 46 years), with longer disease (8.5 vs. 2.0 years), more hypertension (35% vs. 8%) and a narrower CIMT range; their CIMT may reflect traditional risk factors rather than autonomic dysfunction.
- **Disease activity (Fig. 2D):** in IIM, lower ln(RMSSD) was associated with higher MYOACT (β=−0.04, p=0.022).
- **CV risk scores (Fig. 3):** no association with SCORE, SCORE2, mSCORE, PWV or ABI. However, HRV relates to CIMT in dcSSc and IIM.

Figure 3. HRV relates to CIMT, not to conventional risk scores

Standardized β (95% CI), linear regression adjusted for age and sex; filled symbols p<0.05.



Conclusions

- **Vagal activity is reduced in SSc and, numerically, in IIM, compared to HC.**
- **Lower HRV relates to subclinical atherosclerosis in dcSSc and IIM and to disease activity in IIM; in lcSSc, traditional risk factors may prevail.**
- **HRV is unrelated to conventional CV risk scores and may capture CV risk not reflected in current scoring algorithms.**

Abbreviations: ABI - ankle-brachial index; ASyS - antisynthetase syndrome; CI - confidence interval; CIMT - carotid intima-media thickness; CV - cardiovascular; dcSSc/lcSSc - diffuse/limited cutaneous SSc; DM – dermatomyositis; ECG - electrocardiogram; HC - healthy controls; HRV - heart rate variability; IIM - idiopathic inflammatory myopathies; IMNM - immune-mediated necrotizing myopathy; IQR - interquartile range; mSCORE - modified SCORE = SCORE x1.5 ; MYOACT - Myositis Disease Activity Assessment Visual Analogue Scales; PM - polymyositis; PWV - pulse wave velocity; RMSSD - root mean square of successive differences between normal heartbeats; SCORE/SCORE2 - Systematic COronary Risk Evaluation (2); SD - standard deviation; SSc - systemic sclerosis; US - ultrasound.

Reference: 1. Tegegne BS, et al. Reference values of heart rate variability from 10-second resting electrocardiograms: the Lifelines Cohort Study. Eur J Prev Cardiol 2020;27:2191-4.

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